



Small Magellanic Cloud

The Small Magellanic Cloud (left) and the globular cluster 47 Tucanae (right) seem the same distance from us. In reality, the galaxy, which is 10,000 light-years across, is 210,000 light-years away, while 47 Tucanae is just 13,400 light-years away.

to Earth. It appears large in the sky and covers a third of the apparent width of the Moon. As a result, its light is spread out, so it is difficult to spot. Binoculars will locate its pale-gray light. Close by is the bright star Fomalhaut in Piscis Austrinus.

Altair (Aquila), Vega (Lyra), and Deneb (Cygnus) form the Summer Triangle and are still visible to the northwest. September offers the last chance of seeing the triangle before Vega and, later, Deneb move below the horizon. Take a last look at Albireo, in the beak of the swan. It is a celebrated double that is clearly separated through a telescope.

The path of the Milky Way is seen in the western sky. Sagittarius remains high; below is Scorpius. Lowest of all are Centaurus and Crux, which are disappearing below the southwestern horizon. The blue-white star Achernar is well placed to the southeast. At magnitude 0.45, it is the ninth-brightest of all. Achernar marks one end of the mythical river Eridanus. Its name, Arabic in origin, means “river’s end.”

To the right of Achernar is Tucana, representing the tropical bird, the toucan. It is home to the globular cluster 47 Tucanae (NGC 104). Close by is an elongated patch of light seven times wider than the apparent width of the full Moon. This is the Small Magellanic Cloud—a small irregular galaxy that is a companion to the Milky Way. When viewed with binoculars, this naked-eye object reveals some of its star clusters and nebulae.



Helix Nebula

The color and complexity of the Helix Nebula in Aquarius are seen only in images such as this one. The shells of gas, which are about 15 light-years across, are material expelled by a central dying star. The nebula is about 300 light-years away.

SOUTHERN LATITUDES

Capricornus and Aquarius are overhead. Neither has especially bright stars, but Aquarius in particular has objects of note. The brightest star in Capricornus, Alpha (α), called Algedi, is a double star. Sharp eyesight or binoculars show a giant of magnitude 3.6 and, six times more distant, a supergiant of magnitude 4.3.

Aquarius, the water carrier, includes M2, a globular cluster that is easily seen as a fuzzy star through binoculars, and a notable planetary nebula. The Helix Nebula (NGC 7293) is believed to be the closest planetary nebula

